

Jayant Dulani

Software Engineer | Full Stack Developer

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Profile Summary

Software Engineer and Full-Stack Developer with experience building scalable backend systems, automation workflows, and distributed applications using Python, Java, C++, JavaScript, and SQL. Skilled in API development, cloud technologies, Docker, Kubernetes, CI/CD, and database management. Proven ability to optimize system performance, automate business processes, and deliver reliable software solutions in agile environments. Strong foundation in software engineering, problem-solving, and system design.

Technical Skills

Languages: C++, C, Python, Lua, Java, SQL, JavaScript, HTML/CSS

Frameworks & Tools: Git/GitHub, Responsive Web Design, Semantic HTML, Cross-Client Email Testing (Litmus), QA Tools, Email Marketing Platforms (Mailchimp, Campaign Monitor, Salesforce Marketing Cloud)

Cloud & Databases: PostgreSQL, MySQL, SQL Server, AWS

Practices: Industrial Automation, IoT Systems, Backend Development, Distributed Systems, Full SDLC, Agile/Scrum, CI/CD, Containerization (Docker/Kubernetes), System Programming, Data Flow APIs

Soft Skills: Collaboration, Problem-Solving, Analytical Thinking, Clear Communication, Learning, Team Engagement

Work Experience

Jr. Software Engineer

Feb 2026 – Present

Aavya LLC

- Designed and maintained automation workflows using Python and n8n, reducing manual operational workload by 40% and improving process efficiency.
- Developed scalable API integrations with databases and internal systems, enabling reliable data synchronization and seamless workflow interoperability.
- Diagnosed and resolved workflow failures and performance bottlenecks while collaborating with cross-functional teams to deploy automation solutions and document 10+ production workflows.

Undergraduate Teaching Assistant (DSA)

Jan 2025 – Dec 2025

Virginia Tech

- Reviewed and debugged Java and Python programs in Linux, reinforcing cybersecurity, error detection, and IT audit readiness practices.
- Led review sessions for 200+ students, emphasizing training & development, technical support in fast-paced environments, and documentation aligned with business technology practices.

Software Development Intern

May 2024 – Aug 2024

A Round Entertainment

- Developed scalable backend services using Java, Node.js, and C++, building microservices, RESTful APIs, and secure system integrations to support business applications.
- Integrated APIs with Angular dashboards to deliver data analytics and visualization solutions, enabling improved operational insights and decision-making.
- Implemented CI/CD pipelines using GitHub Actions, Jenkins, Maven, and Gradle, while validating Java and Python applications through automated testing to enhance deployment efficiency and system reliability.

Projects

SkipList-Quadtree (Java) | GitHub

February 2024

- Designed and implemented a hybrid SkipList-Quadtree indexing engine for spatial search and 2D range queries.
- Optimized query latency by 30% using hierarchical indexing, memory-local layouts, and advanced data structures.
- Handled dynamic updates with tight memory layout and pointer-based structures, optimizing locality and performance.

Personal Server (Python, C)

December 2024

- Built a scalable, fault-tolerant HTTP server in C/Python with epoll-based concurrency, applying system reliability, event-driven architecture, and debugging at OS and application layers.
- Configured and deployed backend services in C/Python with Docker and PostgreSQL, automating integration workflows and validating CI/CD pipelines.

Personal Terminal ©

September 2024

- Built a POSIX-compliant shell and simulated production-grade workflows with Docker Compose and PostgreSQL, streamlining configuration, testing, and validation of integration pipelines.
- Implemented process scheduling, I/O management, and signal handling with a strong emphasis on memory safety.
- Simulated production-grade backend workflows using Docker Compose and PostgreSQL to validate deployment pipelines.

GPU Stereo Depth & Object Detection Pipeline (C++, CUDA, OpenMP, Python)

December 2025

- Built a real-time autonomous perception system deriving depth and detecting obstacles from stereo cameras alone, eliminating LiDAR and proving that low-cost edge hardware can replicate sensor-rich AV stacks.
- Designed the full pipeline end-to-end: SGBM stereo disparity, WLS filtering, temporal smoothing, and Sobel edge detection fused with YOLO11x object detection on synchronized OAK-D Lite feeds in C++ and Python.
- Parallelized across serial CPU (Raspberry Pi), OpenMP, CUDA GPU, and CuPy live-video paths; achieved 1,380× speedup (13.79 ms → 0.010 ms at 2048×2048) and 135 FPS live on edge hardware.
- Validated metric depth at $R^2=0.9956$ with sub-10% error across a 15-point ground-truth simulation; root-caused a silent WLS scaling bug between float disparity and OpenCV fixed-point format that corrupted all downstream depth measurements.

Education

B.S. in Computer Science — Virginia Tech

GPA: 3.7 | Dean's List with Distinction

Awards & Achievements

- **HackViolet (Hackathon) — Honorable Mention:** Recognized for a fraud detection system using AI/ML and blockchain logging, integrating Java/React features, automated testing workflows, and secure data handling.